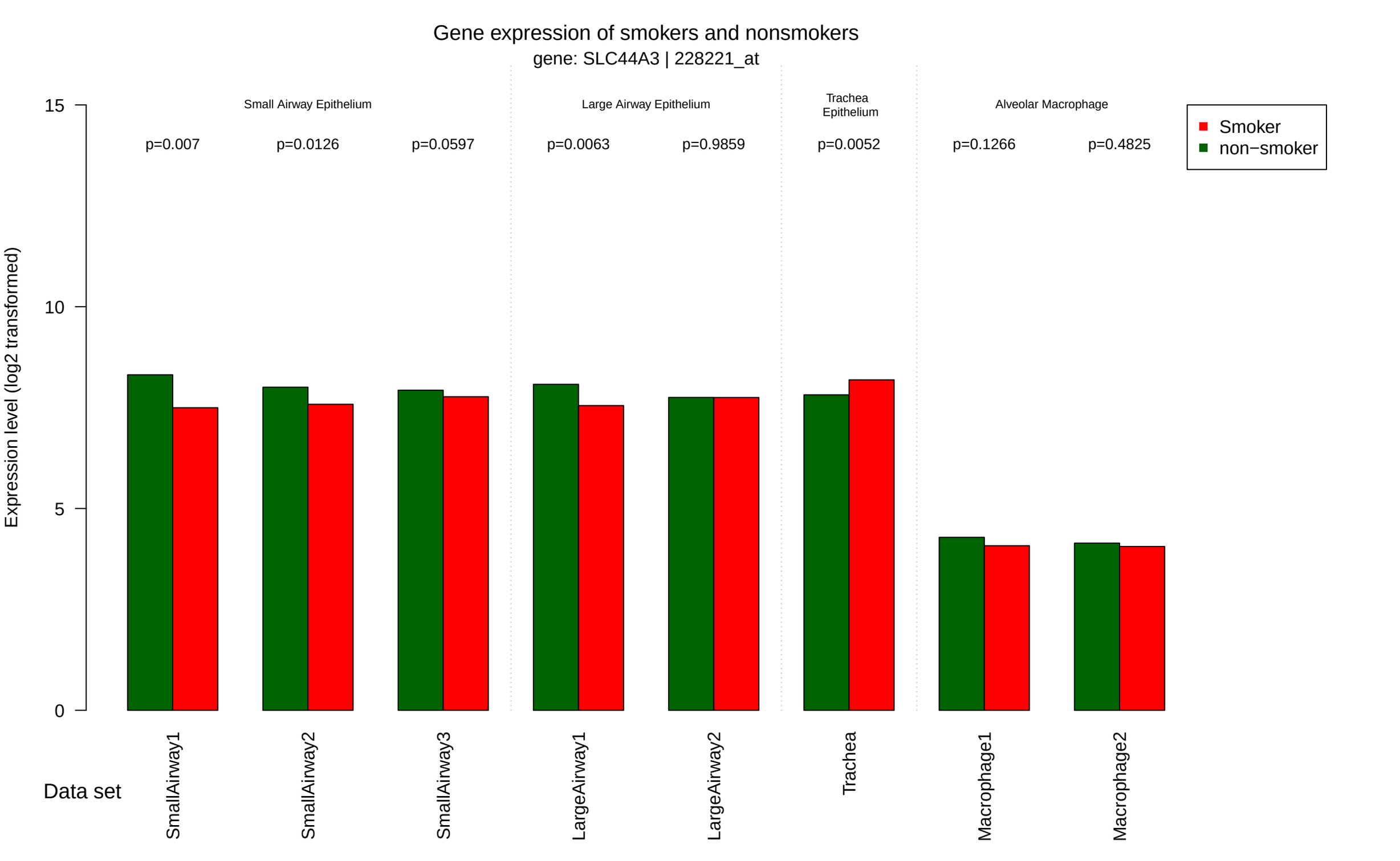
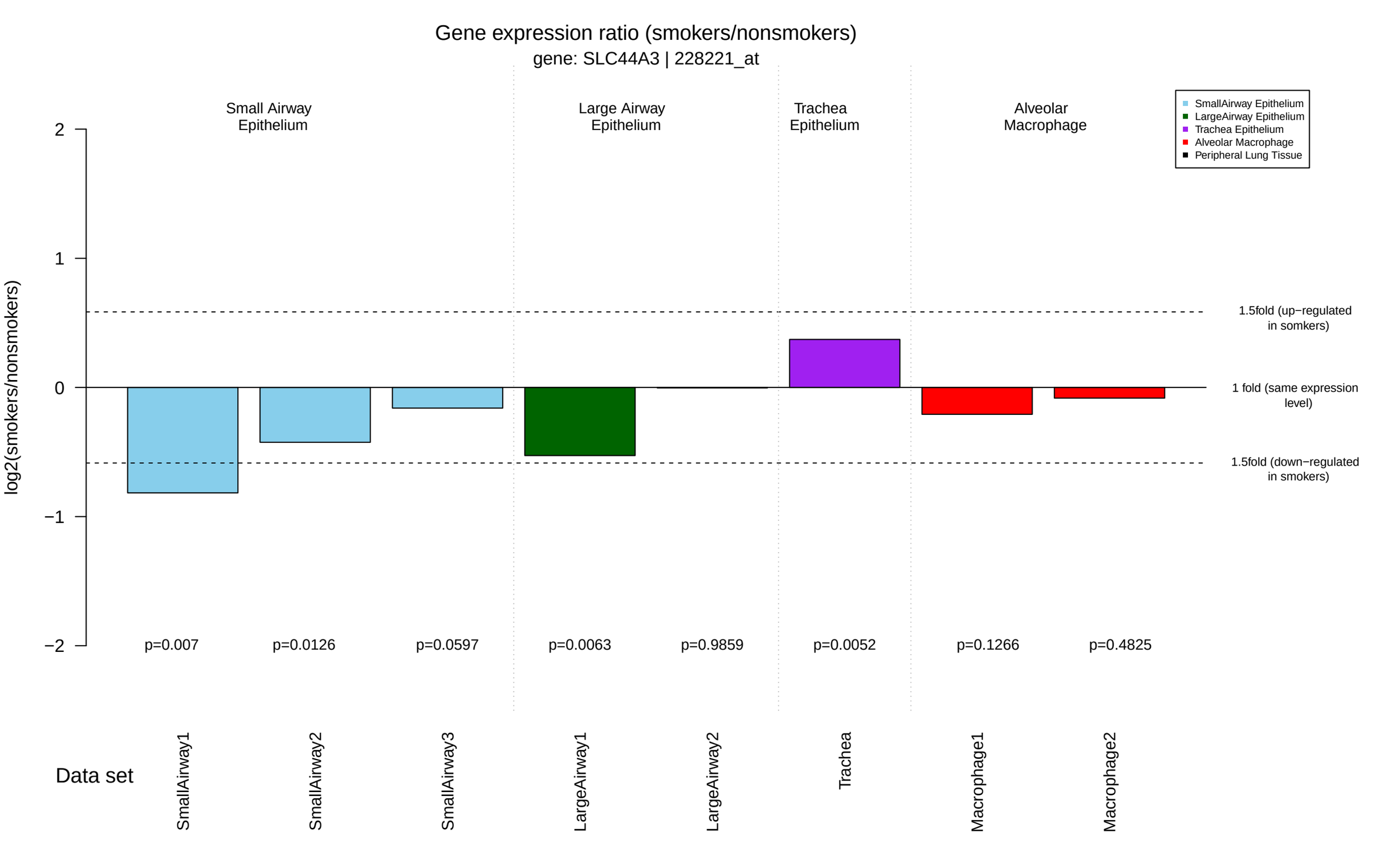
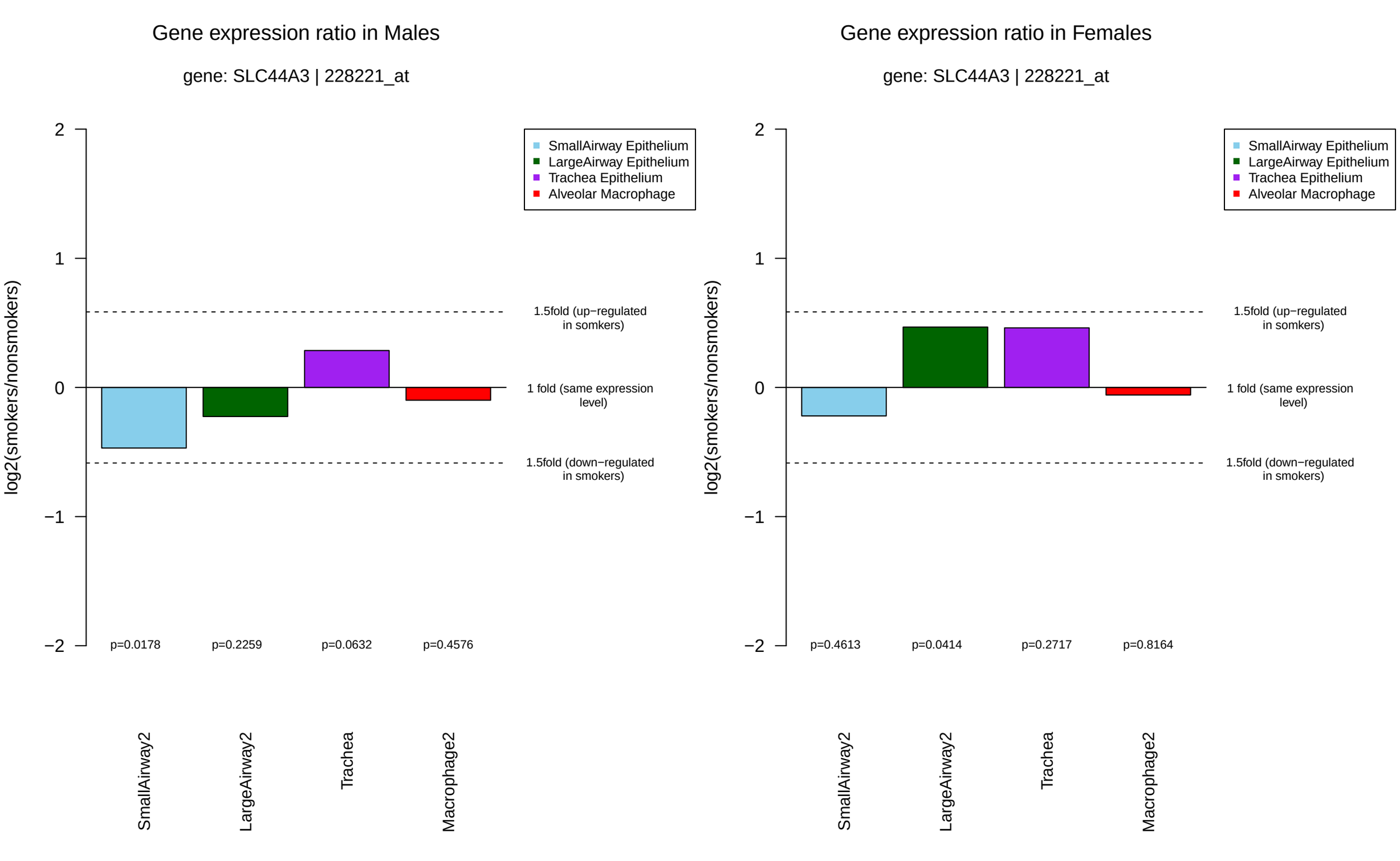
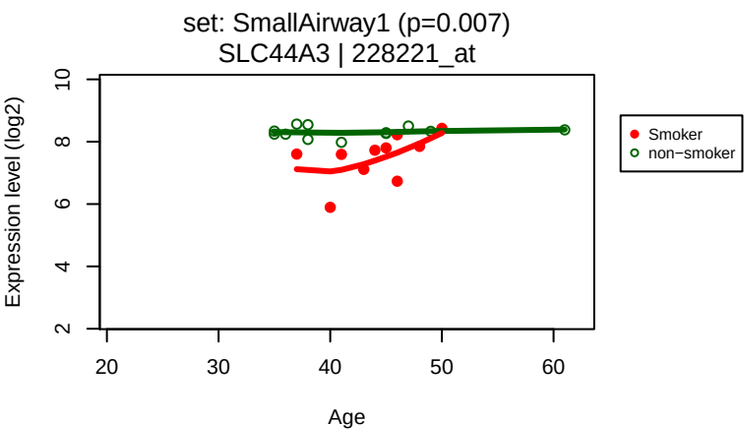
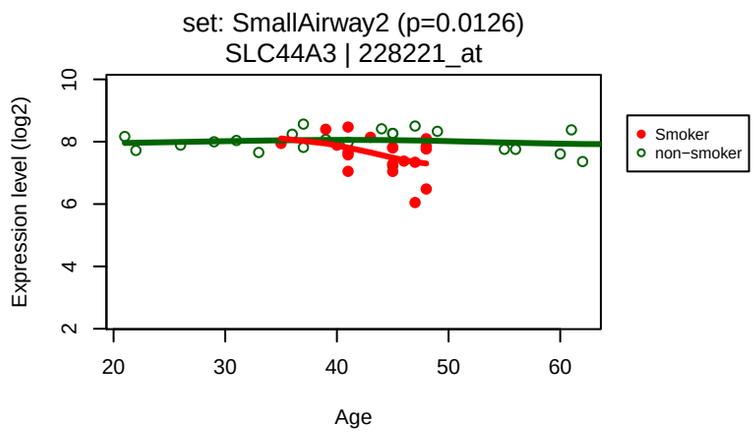
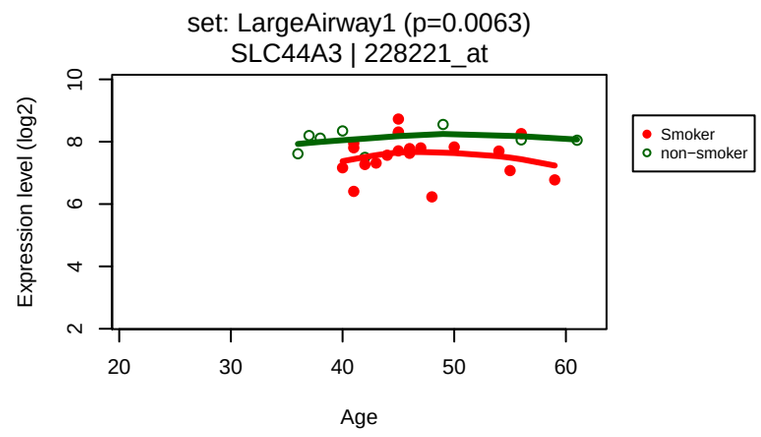

Gene expression of smokers and nonsmokers
gene: SLC44A3 | 228221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.007
p=0.0126
p=0.0597
p=0.0063
p=0.9859
p=0.0052
p=0.1266
p=0.4825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

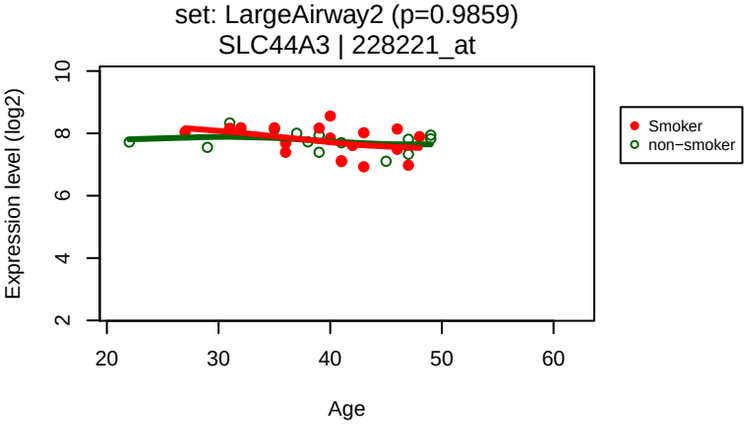

Gene expression ratio (smokers/nonsmokers)
gene: SLC44A3 | 228221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.007
p=0.0126
p=0.0597
p=0.0063
p=0.9859
p=0.0052
p=0.1266
p=0.4825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

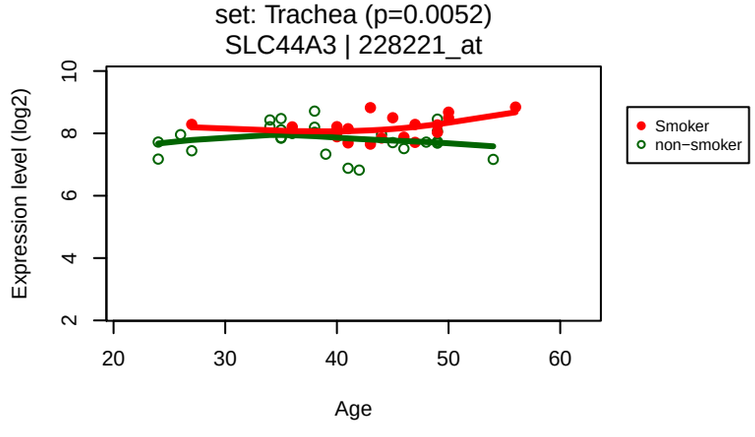

Gene expression ratio in Males
gene: SLC44A3 | 228221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=0.2259
p=0.0632
p=0.4576
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC44A3 | 228221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4613
p=0.0414
p=0.2717
p=0.8164
SmallAirway2
LargeAirway2
Trachea
Macrophage2

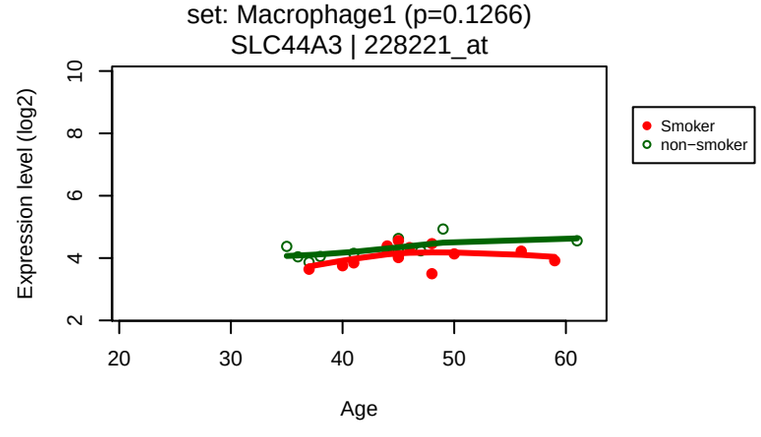

set: SmallAirway1 (p=0.007)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker

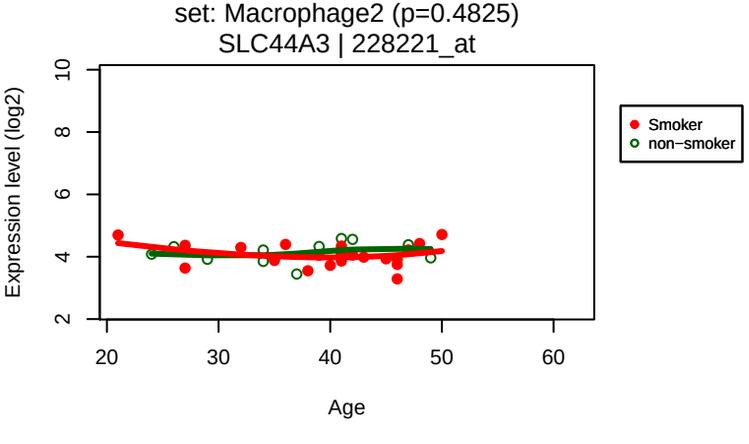

set: SmallAirway2 (p=0.0126)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker

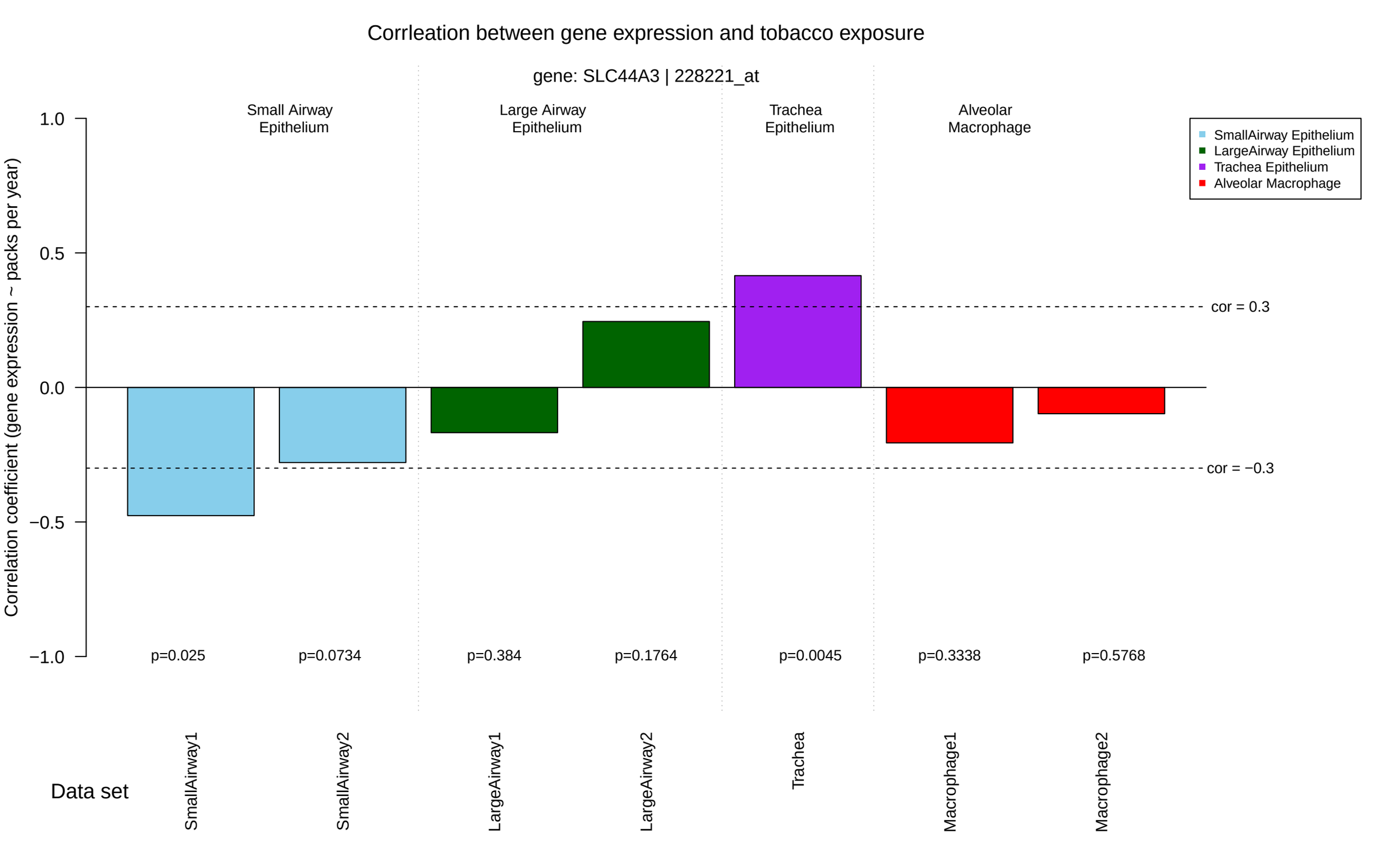

set: LargeAirway1 (p=0.0063)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9859)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0052)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1266)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4825)
SLC44A3 | 228221_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC44A3 | 228221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.025
p=0.0734
p=0.384
p=0.1764
p=0.0045
p=0.3338
p=0.5768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2